

Rahul Lanka

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Professional Summary

Software Engineer with 3+ years of experience designing and delivering scalable full-stack applications using React, TypeScript, JavaScript, Node.js, Python, Supabase, and PostgreSQL. Experienced in database migrations, offline-first synchronization, role-based access control, notification systems, workflow automation, performance optimization, and LLM API integration. Delivered production systems supporting 500+ schools, reducing data-fetch latency by **40%** and lesson load time by **60%**.

Experience

Software Engineer, Chimple Learning

Apr 2023 – Present

Full Stack Development – React, TypeScript, Node.js, Supabase, Firebase

Bengaluru, India

- Architected an end-to-end Firebase-to-Supabase backend migration, including PostgreSQL data modeling, application integration, and offline synchronization, reducing data-fetch latency by **40%** across 500+ schools.
- Built Ops Mode with React, TypeScript, database workflows, API integration, and validated Excel uploads to automate onboarding of schools, classes, and students, reducing manual work by **90%**.
- Designed Supabase Row Level Security policies and authentication-based access controls to secure multi-role application data.
- Developed offline-first learning workflows using Supabase caching and synchronization while optimizing pagination, lazy loading, and rendering to reduce lesson load time by **60%**.
- Automated WhatsApp activity-report generation using Canvas APIs, backend data workflows, and scheduled notifications for student monitoring.
- Improved stability on 1 GB RAM Android devices, introduced Storybook visual testing, and supported releases through code reviews, pull-request validation, and GitHub Actions CI/CD.

QA and Automation Intern, Enhops

Jul 2022 – Apr 2023

Functional, Regression, API, and End-to-End Testing

Hyderabad, India

- Designed and executed 100+ functional, regression, and end-to-end test scenarios, achieving **95% functional coverage** across core application flows.
- Managed defect tracking, retesting, and release validation across Agile sprint cycles to improve application reliability.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, Java, SQL

Frontend: React.js, Redux Toolkit, HTML5, CSS3, Tailwind CSS, Storybook

Backend: Node.js, FastAPI, REST APIs, Authentication, RBAC, Supabase, Firebase

Databases: PostgreSQL, Firebase Realtime Database

AI and LLM: Gemini API, LangChain, RAG, Prompt Engineering, Embeddings, Vector Search

Testing and DevOps: Jest, Postman, Git, GitHub Actions, Docker, CI/CD

Cloud and Tools: AWS, GCP, Vercel, Jira, Confluence, Android Studio

Core Concepts: Data Structures and Algorithms, System Design, Offline-First Architecture, Performance Optimization, Agile Scrum

Projects

Ops Automation System

React, TypeScript, Node.js, Supabase, PostgreSQL

- Built a full stack onboarding platform with React interfaces, validated Excel processing, API integration, PostgreSQL-backed workflows, authentication, and role-based access for schools and students.

LANAI – AI-Powered Learning Platform

TypeScript, Python, LangChain, LLM APIs, RAG

- Contributed to an AI-powered educational platform that uses LLM-based generation and retrieval workflows to transform curriculum content into interactive learning experiences.

Photography Portfolio and Client Gallery (Live)

JavaScript, HTML5, CSS3, Supabase

- Built a responsive full-stack photography platform with Supabase authentication, role-based admin and client access, private event galleries, photo management, and customer-review moderation.

Education

M.Sc. in Computer Science, Government Arts College, Rajahmundry

Mar 2021 | GPA: 8.91

B.Sc. in Computer Science, D.N.R. Degree College, Bhimavaram

Mar 2019 | GPA: 7.67

Achievements

Top 2% in AccioJob Week of Code | Reduced application load time by 60% | Automated onboarding work by 90%